



Dayananda Sagar University

School of Engineering

Department of Computer Science and Technology



RunaGen AI: LLM-Assisted ETL for Role-Based Resume Analytics

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INTRODUCTION

In today's competitive job market, students and job seekers struggle to choose suitable career roles, identify required skills, and prepare effectively for interviews. Existing platforms provide fragmented and generic guidance, leading to confusion and reduced employability.

RunaGen AI is an AI-powered career guidance system that uses intelligent analysis and LLM-based personalization to provide clear, role-specific career and skill guidance.



PROJECT FOCUS

- ETL pipeline development
- Data warehousing and transformations
- Career analytics using real job market data
- Visualization using Power BI / Tableau



DATASET & DATASET COLLECTION

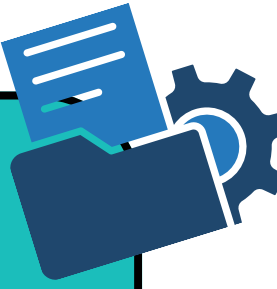
DATASETS USED

- Job Postings: Adzuna API (free tier)
- Skills Taxonomy: ESCO API
- Occupation Data: O*NET Web Services
- Salary Data: Adzuna API
- Supplementary Data: GitHub public datasets



DATASET COLLECTION (ETL)

- APIs are accessed using Python scripts and with Apache Workflow
- Data is collected daily / weekly
- Incremental loads avoid duplication
- Raw data is stored in staging tables



ETL FLOW:

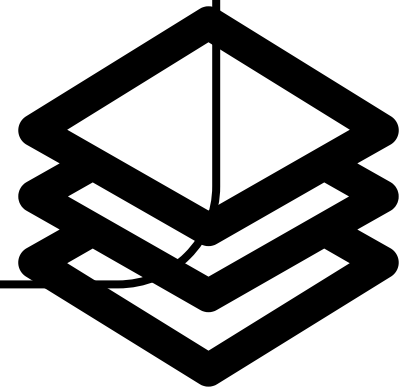




DATASET ATTRIBUTES & SIZE

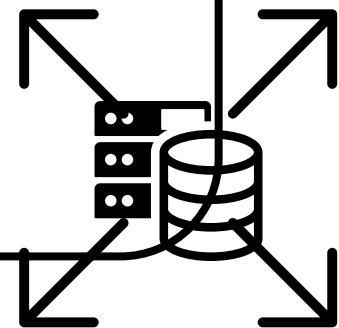
KEY DATASET ATTRIBUTES

- Job Title, Company, Location
- Required Skills (standardized)
- Salary Min / Max
- Experience & Education Level
- Occupation Code
- Extraction Timestamp



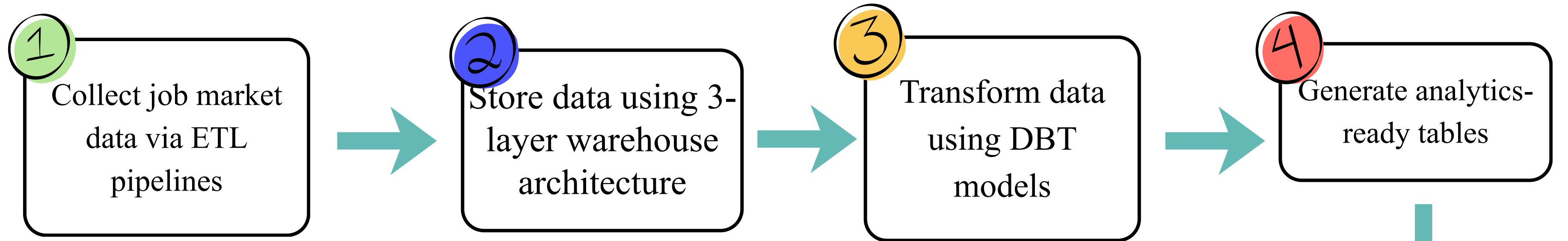
SIZE OF DATASET

- Job postings: ~30,000 records/month
- Skills taxonomy: ~13,000 skills
- Total processed data: ~270,000 records
- Storage size: < 1 GB



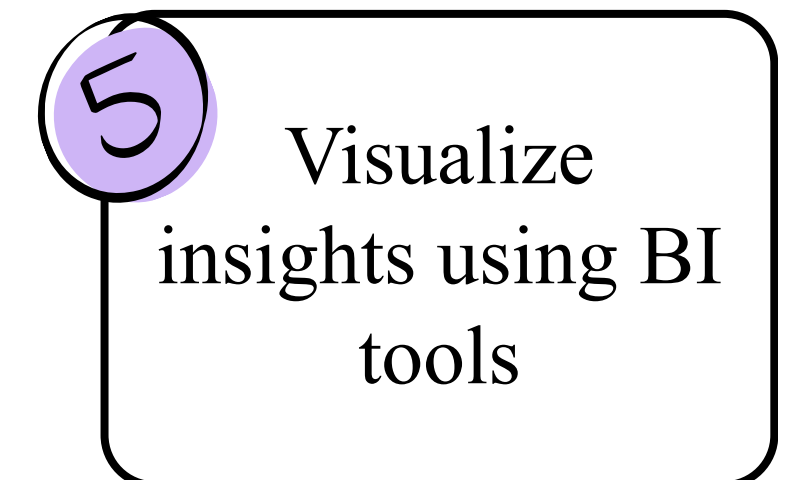


PROPOSED METHODOLOGY



NOVEL MODEL DEVELOPMENT

1. Career Trajectory Probability Model:
 - Predicts role transitions using skill overlap and demand
2. Skill Gap Prioritisation Model:
 - Ranks skills based on demand and salary impact
3. Salary Insight Model:
 - Estimates salary ranges for roles and skills
 - These models are data-driven and reproducible.





ANALYSIS USING POWER BI / TABLEAU

ANALYTICS PERFORMED

- Career transition paths between roles
- Skill demand vs salary premium analysis
- Salary distribution by role and location
- Job market trends over time



DASHBOARDS

- Career Trajectory Dashboard
- Skill Gap Analysis Dashboard
- Salary Insights Dashboard
- Job Market Trends Dashboard



OUTCOME

- Clear career paths backed by data
- Identification of high-value skills
- Market-aligned salary expectations

An aerial photograph of a long, multi-lane highway bridge spanning a body of water. The water is a vibrant teal color with visible ripples. The bridge has several lanes in each direction, with white lane markings. Several vehicles, including white and green trucks, are visible traveling across the bridge. The text "Thank You" is overlaid in the center of the image in a white, serif font.

Thank You